Effect of Walking Fitness on Brain Health

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Demo R Pipeline: https://github.com/Edward932/Brain-Age-Gap-Demo-Project

Abstract

Brain age gap (BAG) is a neuroimaging-derived marker of accelerated brain aging. BAG is calculated by training a machine learning (ML) model to predict age based on MRI scans (ie. BAG = model estimation - true age). Higher BAG is linked to higher risks of cognitive decline, dementia, and multiple sclerosis (Zhang et al, 2025).

This project aims to test whether higher fitness levels are associated with lower BAG scores. We utilized MRI measurement data to calculate BAG scores then assessed the relation of two-minute walk scores (distance a patient can walk in two minutes) with BAG. We found that each additional 100 feet walked was associated with a 1.61-year decrease in calibrated BAG (95% CI: 1.05 to 2.16 years lower), holding other covariates constant.

Methods

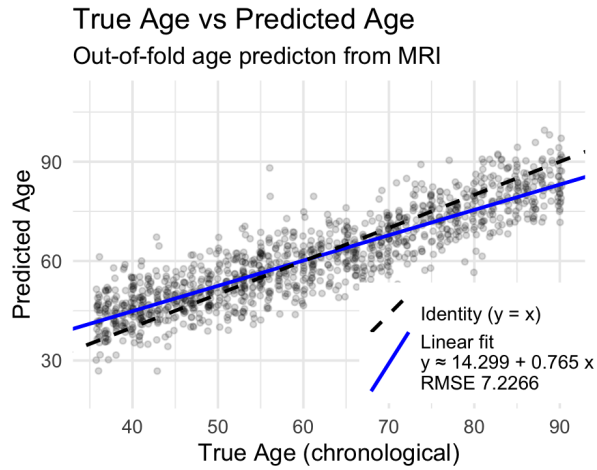
We utilized data from the Aging Adult Brain Connectome (AABC) Consortium. We utilized MRI measurement data consisting of ~1400 measurements that summarize each MRI scan. Two-minute walk scores along with potential confounders (sex, age, BMI, education level, and blood pressure) were also gathered from the AABC data set.

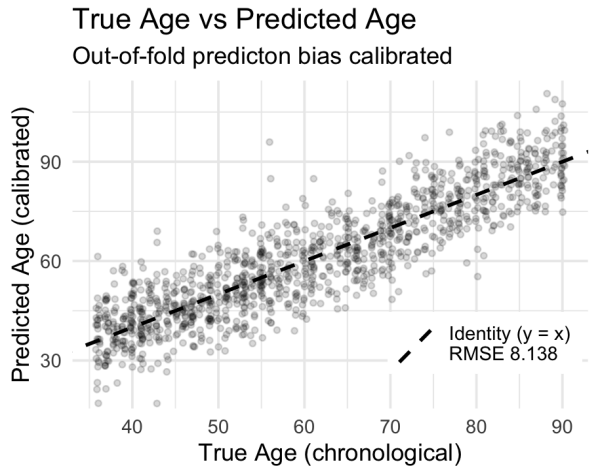
Two ML models were trained to predict brain age. The first model was an Elastic Net with an alpha selected by a brute force method. The second model was XGBoost with 9 hyperparameters selected by a Bayesian-optimization-like procedure via mlr3mbo. All predictions are out-of-fold (cross-validated) to avoid leakage.

Both models required an age bias calibration because predictions regress toward the mean age which is a common problem even in state-of-the-art models (Kim et al, 2025). Calibration is to remove age-dependent bias in BAG but may not improve RMSE. Model performance was very similar (table below), so Elastic Net predicted BAG was selected for downstream analysis because it is a more interpretable and stable model.

	RMSE	RMSE - age bias calibrated
Elastic Net	7.227	8.138
XGBoost	7.219	8.097

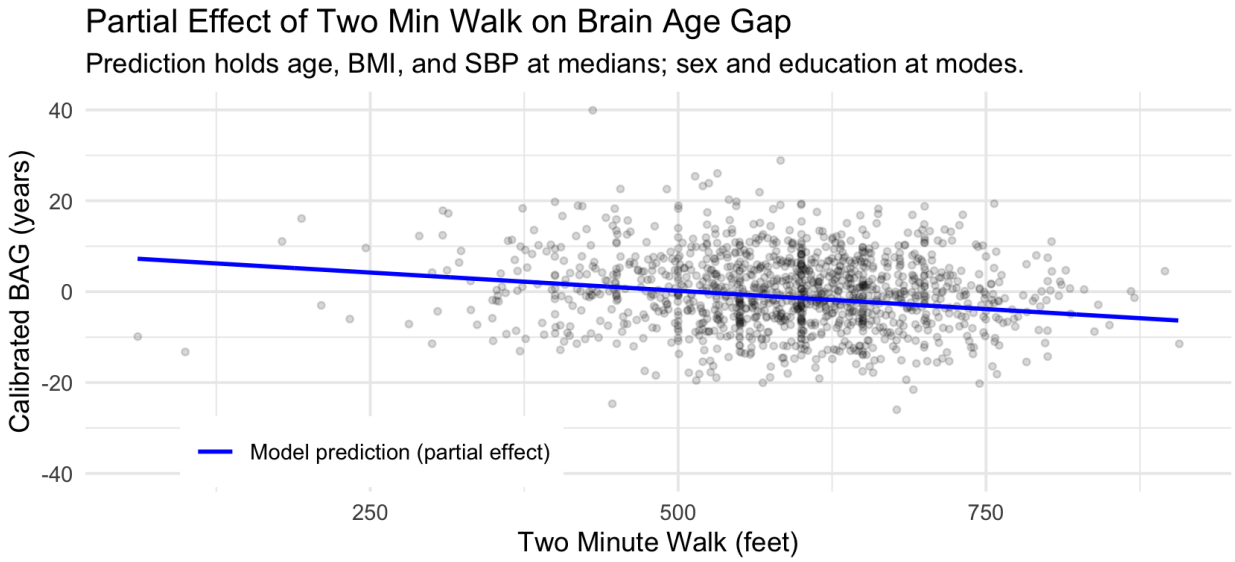
The graphs below show Elastic Net brain age predictions before and after calibration.





Results

We estimated the association between walking endurance and calibrated BAG using multivariable linear regression, adjusting for age, sex, BMI, education, and blood pressure. We found that a greater two-minute walk distance was associated with a lower BAG (ie younger looking brain). Each additional 100 feet walked was associated with a 1.61-year decrease in calibrated BAG (95% CI: 1.05 to 2.16 years lower). Moving from the 25th percentile (528.9 ft) to the 75th percentile (652.7 ft) corresponded to an ~ 2.0-year decrease in BAG holding other covariates constant.



References

1. Zhang, R., Yi, F., Mao, H. *et al.* Brain age gap as a predictive biomarker that links aging, lifestyle, and neuropsychiatric health. *Commun Med* 5, 441 (2025). https://doi.org/10.1038/s43856-025-01100-5

2. Kim, H., Park, S., Seo, S.W. *et al.* A novel deep learning-based brain age prediction framework for routine clinical MRI scans. *npj Aging* 11, 70 (2025). https://doi.org/10.1038/s41514-025-00260-x